BetaQuest

An Algorithmic Rock Climbing Route Finder

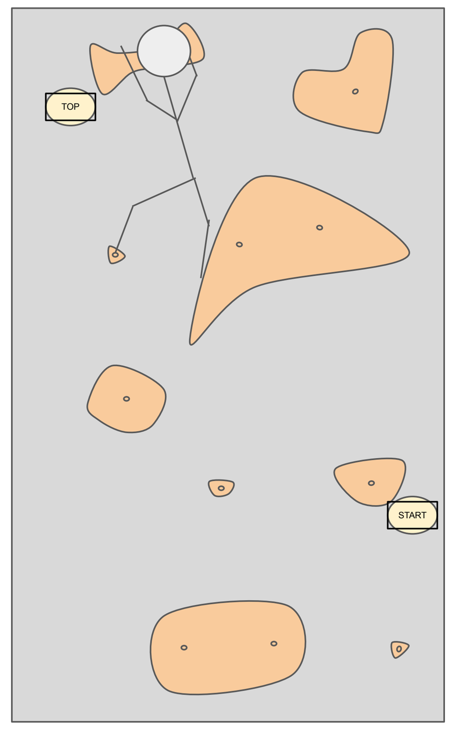
Summer Alwood

Dr Yilmaz; Systems Lab 2026

Introduction

Currently, there are no existing algorithmic beta-finding resources for climbers. This limits their capacity to approach routes for which they have no clue where to start. Thus, the goal of this research is to create a program that can generate an acceptable "beta" (a means of getting up a route) to provide a guideline for climbers trying to approach new climbs. Additionally, we aim to explore the intricacies of human limb movement, accounting for balance, reach, and more.

Because this project is so complex, it was necessary to make several simplifications including: (1) assuming walls are completely flat, (2) representing holds as boxes, (3) and representing position using the position of the four limbs.



Full Paper

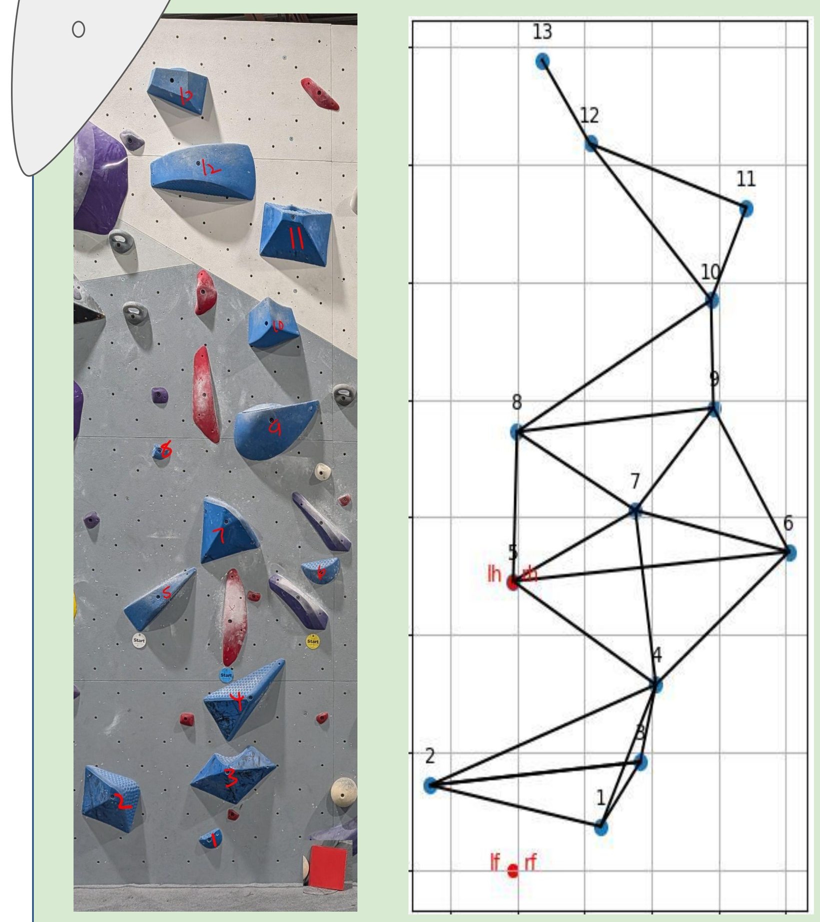


Figure 1: Climb to Graph

Quality

Quality evaluation for a single move was done based on:

1. the estimated strength of the destination hold
2. the distance between the current and destination holds
3. Several rewards and penalties were applied to account for more complex relationships, weighted against one another as indicated (*Figure 3*):

Rewards	Penalties
(20) taking foot off floor	(10) going down
(5) topping out	(6) Right limb further left than left counterpart
	(5) unbalanced *scaled to (hand_x-foot_x)**2*
	(3) (over)extension *"amount" extended = sum((32/3)*((dist-0.5)**4 for each h/h, f/f, h/f pair)*
	(2) matching on a hold *"quality" of match hold = 1/(10*size_x+1))*

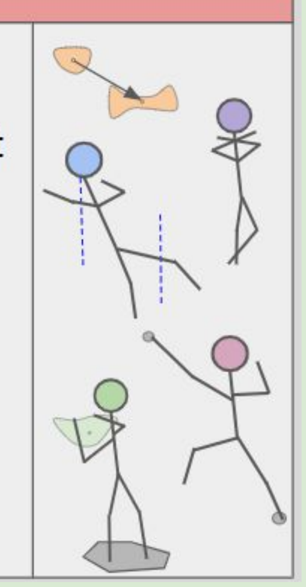


Figure 3: Reward and Penalties applied

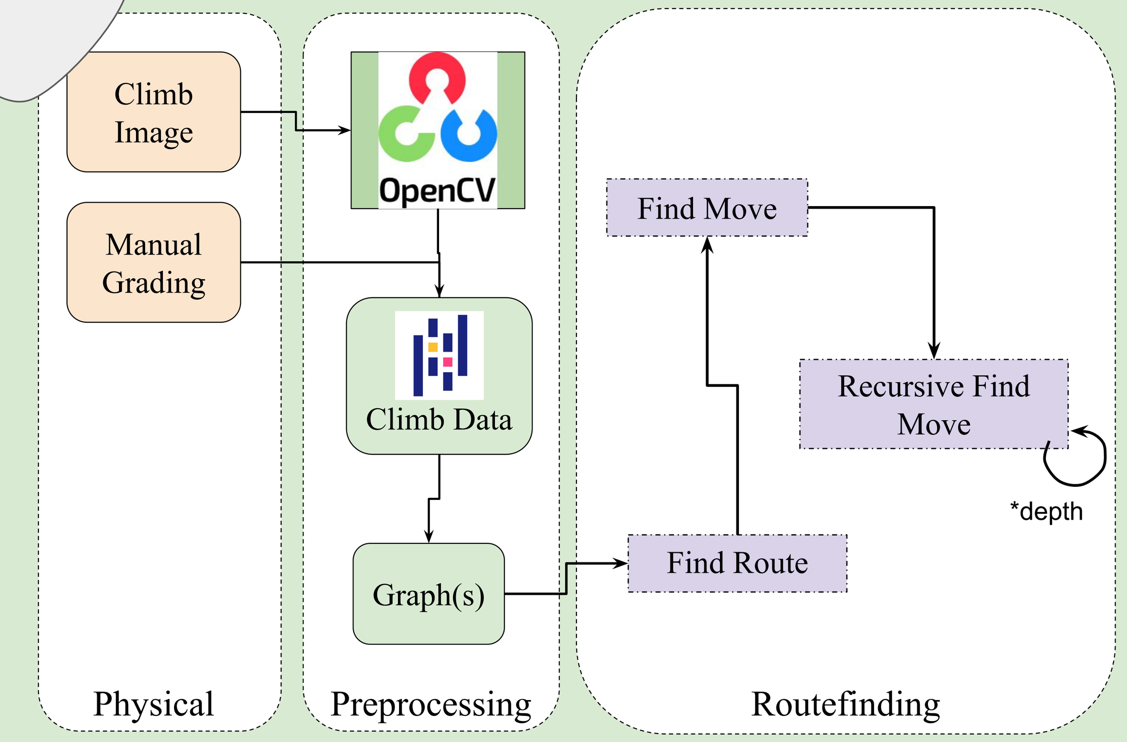


Figure 2: System Architecture Diagram

Methods

Data

Data on indoor rock climbs [holds] was collected at Sportrock Alexandria. This data was processed alongside an image of its respective route and used to create graphs representing climbs (*Figure 1*).

Route Finding Pseudocode

(*Figure 2*)

- position = initial position
- while we haven't reached the top:
  - find the [k] best moves from the current position
  - for each move m:
    - find the quality of every chain of length [n] beginning with m
    - store the quality of the best chain for m
  - choose and store the move with the highest quality chain
  - update position

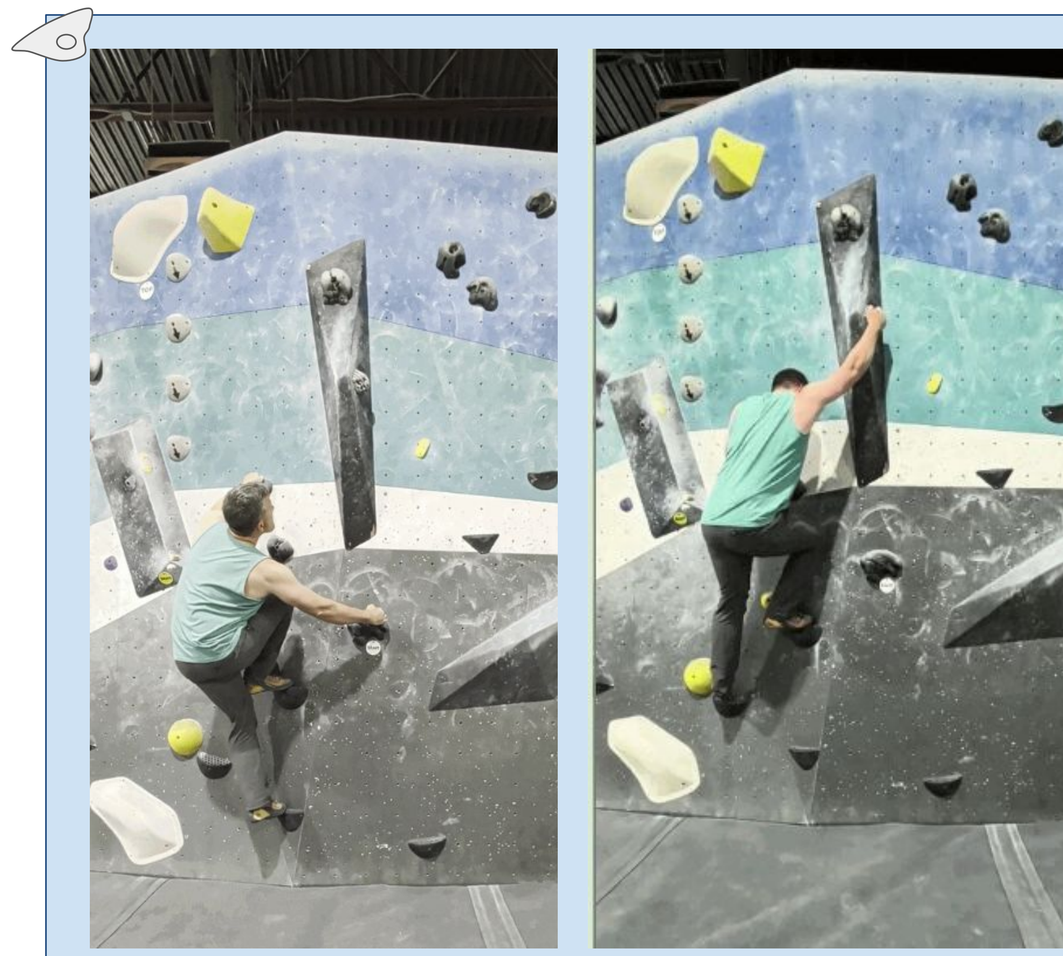


Figure 4: Climber Attempts Route

Results

When evaluated against human generated beta and human feedback on 3 climbs of different types, connectedness, and difficulty, the algorithm was able to produce a completable route with no dead ends or impossible moves.

According to climber feedback, the output was a "good beginner beta" that "felt balanced." However, though the route is output both as text and visual, it was difficult to remember while on the wall. This limited the algorithm's effectiveness, as it led to movement being stilted.

Scan for Full Demo!

Conclusion & Future Work

Although the final product is able to provide a route, it remains imperfect; several moves are unnatural, backwards, or unnecessary. This research could be extended in many ways, including: accounting for non-flat walls, utilization of reinforcement learning with human feedback (RLHF) to optimize quality calculation, and allowing for move chaining or more complex moves. Additionally, it is extremely unlikely that a perfectly generalizable solution to this problem will be found; there is simply too much variance between climbs. It may benefit future researchers to adjust evaluation methods based on qualities of a climb as a whole.